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LOWER PLASTIC MEMBER, END COVER ASSEMBLY, ENERGY STORAGE APPARATUS, AND POWER CONSUMING DEVICE

Abstract

A lower plastic member, an end cover assembly, an energy storage apparatus, and a power consuming device are provided. The lower plastic member includes a body and a suction portion. The body has a first surface and a second surface. The first surface is configured to be positioned at one side of the body facing towards an upper cover. The suction portion is disposed on the first surface. The suction portion includes multiple first ribs and multiple second ribs. The multiple first ribs are parallel to and spaced apart from one another. The multiple second ribs are parallel to and spaced apart from one another. The multiple first ribs and the multiple second ribs are disposed in a crisscross pattern. The multiple first ribs and the multiple second ribs cooperatively define multiple recesses. Each of the multiple recesses is recessed from the first surface towards the second surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202410253862.9, filed Mar. 6, 2024, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure relates to the field of energy storage technology, and in particular to a lower plastic member, an end cover assembly, an energy storage apparatus, and a power consuming device.

BACKGROUND

[0003] Secondary batteries, also known as rechargeable batteries or storage batteries, refer to batteries that can be reused after active materials thereof are reactivated through charging following discharge. With the recyclable characteristics of the secondary battery, the secondary battery gradually becomes a main power source of a power consuming device. With the increasing demand of the secondary battery, people have higher and higher performance requirements for the secondary battery in various aspects, especially for the energy density per unit volume of the battery. The thickness of an end cover assembly of the battery is an important parameter affecting the energy density per unit volume of battery, and an excessively thick end cover assembly may reduce the energy density per unit volume of the battery. At present, the end cover assembly includes a lower plastic member used for insulation between an upper cover and a terminal. In order to improve the energy density per unit volume of the battery, the lower plastic member is usually designed to be very thin.

SUMMARY

[0004] In a first aspect, a lower plastic member for an end cover assembly of an energy storage apparatus is provided in the present disclosure. The lower plastic member includes a body, a suction portion, and an explosion-proof boss. The body has a first surface and a second surface positioned facing away from the first surface. The first surface is configured to be positioned at one side of the body facing towards an upper cover. The suction portion is disposed on the first surface. The suction portion includes multiple first ribs and multiple second ribs. The multiple first ribs are parallel to and spaced apart from one another. The multiple second ribs are parallel to and spaced apart from one another. The multiple first ribs and the multiple second ribs are disposed in a crisscross pattern. The multiple first ribs and the multiple second ribs cooperatively define multiple recesses. Each of the multiple recesses is recessed from the first surface towards the second surface. The explosion-proof boss is disposed at a middle part of the body in a length direction of the body. The explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface. The explosion-proof boss is configured to be aligned with an explosion-proof valve disposed on the upper cover. A midline extending in a width direction of the first surface is corresponding to the explosion-proof boss. There is a spacing distance between the suction portion and the explosion-proof boss. An end surface of the multiple first ribs configured to be positioned facing towards the upper cover is flush with the first surface. An end surface of the multiple second ribs configured to be positioned facing towards the upper cover is flush with the first surface. An orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm. An orthographic projection of each of the multiple recesses on the

first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm. A depth of each of the multiple recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm. The suction portion is implemented as multiple suction portions. The multiple suction portions are symmetrical about the midline extending in the width direction of the first surface. The multiple suction portions are arranged symmetrically about a second midline.

[0005] In a second aspect, an end cover assembly is further provided in the present disclosure. The end cover assembly includes an upper cover and a lower plastic member. The lower plastic member includes a body, a suction portion, and an explosion-proof boss. The body has a first surface and a second surface positioned facing away from the first surface. The first surface of the lower plastic member is positioned facing towards the upper cover. The suction portion is disposed on the first surface. The suction portion includes multiple first ribs and multiple second ribs. The multiple first ribs are parallel to and spaced apart from one another. The multiple second ribs are parallel to and spaced apart from one another. The multiple first ribs and the multiple second ribs are disposed in a crisscross pattern. The multiple first ribs and the multiple second ribs cooperatively define multiple recesses. Each of the multiple recesses is recessed from the first surface towards the second surface. The explosion-proof boss is disposed at a middle part of the body in a length direction of the body. The explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface. The explosion-proof boss is aligned with an explosion-proof valve disposed on the upper cover. A midline extending in a width direction of the first surface is corresponding to the explosion-proof boss. There is a spacing distance between the suction portion and the explosion-proof boss. An end surface of the multiple first ribs positioned facing towards the upper cover is flush with the first surface. An end surface of the multiple second ribs positioned facing towards the upper cover is flush with the first surface. An orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm. An orthographic projection of each of the multiple recesses on the first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm. A depth of each of the multiple recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm. The suction portion is implemented as multiple suction portions. The multiple suction portions are symmetrical about the midline extending in the width direction of the first surface. The multiple suction portions are arranged symmetrically about a second midline.

[0006] In a third aspect, a power consuming device is further provided in the present disclosure. The power consuming device includes a power consuming apparatus and an energy storage apparatus. The energy storage apparatus is configured to supply power to the power consuming apparatus. The energy storage apparatus includes a housing, a cell assembly, and an end cover assembly. The cell assembly is disposed in the housing. The end cover assembly is connected to the housing. The end cover assembly includes an upper cover and a lower plastic member. The lower plastic member includes a body, a suction portion, and an explosion-proof boss. The body has a first surface and a second surface positioned facing away from the first surface. The first surface of the lower plastic member is positioned facing towards the upper cover. The suction portion is disposed on the first surface. The suction portion includes multiple first ribs and multiple second ribs. The multiple first ribs are parallel to and spaced apart from one another. The multiple second ribs are parallel to and spaced apart from one another. The multiple first ribs and the multiple second ribs are disposed in a crisscross pattern. The multiple first ribs and the multiple second ribs cooperatively define multiple recesses. Each of the multiple recesses is recessed from the first surface towards the second surface. The explosion-proof boss is disposed at a middle part of the body in a length direction of the body. The explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface. The explosion-proof boss is aligned with an explosion-proof valve disposed on the upper cover. A midline extending in a width direction of the first surface is corresponding to the explosion-proof boss. There is a spacing distance between the suction portion and the explosion-proof boss. An end surface of the multiple first ribs positioned

facing towards the upper cover is flush with the first surface. An end surface of the multiple second ribs positioned facing towards the upper cover is flush with the first surface. An orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm. An orthographic projection of each of the multiple recesses on the first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm. A depth of each of the multiple recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm. The suction portion is implemented as multiple suction portions. The multiple suction portions are symmetrical about the midline extending in the width direction of the first surface. The multiple suction portions are arranged symmetrically about a second midline.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] To more clearly describe implementations in the present disclosure or technical solutions in related art, the accompanying drawings that need to be used in description of implementations or the related art will be briefly introduced below. Apparently, the accompanying drawings in the following description are only some implementations in the present disclosure, and those of ordinary skill in the art may also obtain other accompanying drawings based on these accompanying drawings without creative effort.

[0008] FIG. 1 is a view of an application scenarios of an energy storage apparatus according to an embodiment.

[0009] FIG. 2 is a perspective view of a lower plastic member according to an embodiment, along with a partial enlarged view thereof.

[0010] FIG. 3 is a perspective view of a lower plastic member according to an embodiment, viewed from another direction.

[0011] FIG. 4 is a schematic view of injection molding of a lower plastic member according to an embodiment.

[0012] FIG. 5 is a schematic view of demolding of a lower plastic member according to an embodiment.

[0013] FIG. 6 is a bottom view of a lower plastic member according to an embodiment.

[0014] FIG. 7 is a top view of a lower plastic member according to an embodiment, along with a partial enlarged view thereof.

[0015] FIG. 8 is a partial cross-sectional view of a lower plastic member according to an embodiment.

[0016] FIG. 9 is an exploded schematic view of an energy storage apparatus according to an embodiment.

[0017] FIG. 10 is a block view of a power consuming device according to an embodiment.

[0018] Description of reference signs of the accompanying drawings: **1000**—energy storage apparatus, **2000**—photovoltaic panel, **3000**—wind turbine, **4000**—power grid, **10**—lower plastic member, **101**—gate portion, **11**—body, **111**—first surface, **112**—second surface, **12**—suction portion, **121**—recess, **122**—first rib, **123**—second rib, **13**—code portion, **14**—terminal boss, **141**—first terminal boss, **142**—first terminal-hole, **143**—first terminal-groove, **144**—second terminal boss, **146**—second terminal-hole, **147**—avoidance notch, **15**—snap-fit, **151**—extension portion, **152**—protrusion, **16**—slider portion, **171**—annular push portion, **172**—first push-point portion, **173**—second push-point portion, **174**—third push-point portion, **18**—explosion-proof boss, **181**—first boss, **182**—third surface, **183**—vent hole, **184**—second boss, **185**—fourth surface, **186**—air-permeable hole, **191**—first recessed platform, **192**—fifth surface, **193**—first leakage hole, **194**—second recessed platform, **195**—sixth surface, **196**—second leakage hole, **197**—first reinforcing rib, **198**—second reinforcing rib, **201**—mold-mating plane, **21**—first mold, **211**—first molding

surface, **212**—first molding groove, **213**—push hole, **22**—second mold, **221**—second molding surface, **222**—second molding groove, **223**—butting portion, **224**—butting surface, **225**—pouring hole, **A1**—first midline, **A2**—second midline, **X**—first direction, **Y**—second direction, **Z**—third direction, **20**—upper cover, **23**—explosion-proof valve, **100**—end cover assembly, **200**—housing, **300**—cell assembly, **1100**—power consuming apparatus, **5000**—power consuming device.

DETAILED DESCRIPTION

[0019] Technical solutions of implementations in the present disclosure will be described clearly and completely below with reference to accompanying drawings in implementations of the present disclosure. Apparently, implementations described herein are merely some implementations, rather than all implementations, of the present disclosure. Based on implementations of the present disclosure, all other implementations obtained by those of ordinary skill in the art without creative effort shall fall within the protection scope of the present disclosure.

[0020] It may be noted that when an assembly is referred to as “fixed to” another assembly, the assembly may be directly positioned on the other assembly or an intermediate assembly may exist therebetween. When an assembly is referred to as “connected to” another assembly, the assembly may be directly connected to the other assembly or an intermediate assembly may exist therebetween simultaneously.

[0021] Unless otherwise defined, all technical and scientific terms used in the present disclosure have the same meaning as commonly understood by those skilled in the art of the present disclosure. The terms used in the detailed description in the present disclosure are for the purpose of describing specific embodiments only and are not intended to limit the present disclosure. The term “and/or” in the present disclosure includes any and all combinations of one or more related listed items.

[0022] The following will describe in detail some implementations of the present disclosure with reference to the accompanying drawings. Various embodiments and features therein may be implemented in any combination with each other without conflict.

[0023] At present, the lower plastic member is manufactured by using an injection molding process. During demolding, the lower plastic member is easily damaged due to pulling of a mold, and the product yield is low.

[0024] The present disclosure aims to provide a lower plastic member, an end cover assembly, an energy storage apparatus, and a power consuming device. Therefore, a problem, that the lower plastic member is easily damaged due to pulling of a mold during demolding, is solved.

[0025] In order to realize the purpose of the present disclosure, the present disclosure provides the following technical solution.

[0026] In a first aspect, a lower plastic member for an end cover assembly of an energy storage apparatus is provided in the present disclosure. The lower plastic member includes a body, a suction portion, and an explosion-proof boss. The body has a first surface and a second surface positioned facing away from the first surface. The first surface is configured to be positioned at one side of the body facing towards an upper cover. The suction portion is disposed on the first surface. The suction portion includes multiple first ribs and multiple second ribs. The multiple first ribs are parallel to and spaced apart from one another. The multiple second ribs are parallel to and spaced apart from one another. The multiple first ribs and the multiple second ribs are disposed in a crisscross pattern. The multiple first ribs and the multiple second ribs cooperatively define multiple recesses. Each of the multiple recesses is recessed from the first surface towards the second surface. The explosion-proof boss is disposed at a middle part of the body in a length direction of the body. The explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface. The explosion-proof boss is configured to be aligned with an explosion-proof valve disposed on the upper cover. A midline extending in a width direction of the first surface is corresponding to the explosion-proof boss. There is a spacing distance between the suction portion and the explosion-proof boss. An end surface of the multiple first ribs configured to be positioned

facing towards the upper cover is flush with the first surface. An end surface of the multiple second ribs configured to be positioned facing towards the upper cover is flush with the first surface. An orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm. An orthographic projection of each of the multiple recesses on the first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm. A depth of each of the multiple recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm. The suction portion is implemented as multiple suction portions. The multiple suction portions are symmetrical about the midline extending in the width direction of the first surface. The multiple suction portions are arranged symmetrically about a second midline.

[0027] In the lower plastic member provided in the present disclosure, the suction portion is provided, and the suction portion is disposed on the first surface and has the recesses recessed from the first surface towards the second surface. During demolding of the mold, the recesses form an effect like a suction cup, so that the suction force between the lower plastic member and a first mold can be increased. A pulling force of a second mold on the lower plastic member is insufficient to separate the lower plastic member from the first mold, thereby avoiding damage to the lower plastic member and improving product yield. The multiple first ribs and the multiple second ribs are disposed to define the multiple recesses. Compared to a single recess with a relatively large size, the multiple dense recesses with relatively small size can have a more stable effect on increasing the suction force with the mold, and have a better effect on avoiding the damage to the lower plastic member and improving the product yield. By setting each dimension of the above suction portion, the suction force between the lower plastic member and the mold can be increased, and the damage to the lower plastic member caused by the bending and deformation of the lower plastic member can be avoided. In addition, the structure is simple, the manufacturing is convenient, and the cost is low. The multiple suction portions symmetrically arranged feature a simple layout, thereby facilitating the mold manufacturing and also facilitating visual recognition during the subsequent assembly of the end cover assembly.

[0028] In an implementation, an end portion of each of the multiple first ribs and the multiple second ribs, configured to be positioned facing towards the upper cover, has a pointed-prism shape.

[0029] In an implementation, at least two of the multiple suction portions are disposed on a same side of the midline extending in the width direction of the first surface, and have different distances from the midline extending in the width direction of the first surface. This arrangement of the multiple suction portions provides a foolproof function for machine vision recognition in automated production lines during the assembly of the end cover assembly, thereby preventing the lower plastic member from being placed incorrectly in a reversed left-right orientation.

[0030] In an implementation, the first surface is provided with a code portion. The code portion has a preset shape. The preset shape includes at least one of a number, a letter, a bar code, or a two-dimensional code. By providing the code portion, which can be recognized with the naked eye or recognized by scanning the code, the product category, production date, manufacturer name, and other characteristics of the lower plastic member can be obtained, thereby facilitating the traceability of the lower plastic member, and the control of the production process and the traceability of quality issues.

[0031] In an implementation, the code portion is implemented as multiple code portions arranged side by side. By arranging the multiple code portions side by side, different kinds of information can be displayed or recognized in one position, which is convenient for tracing various kinds of information.

[0032] In an implementation, any adjacent two of the multiple code portions are spaced apart from each other. During demolding, after the second mold is separated from the first mold and the lower plastic member, the lower plastic member needs to be separated from the first mold. The multiple code portions correspond to multiple replaceable code blocks on the first mold. By providing a gap between any two adjacent code portions, or in other words, between any two adjacent movable

code portions, air can enter the gap during demolding. Therefore, the local negative pressure between the replaceable code blocks and the lower plastic member is avoided, the demolding resistance between the lower plastic member and the first mold is reduced, and the production efficiency is improved.

[0033] In an implementation, the lower plastic member further includes a terminal boss. The terminal boss is connected to the body and protrudes from the first surface. A snap-fit is provided at each of the two sides of the terminal boss in the width direction of the first surface. By providing the snap-fit at each of the two sides of the terminal boss in a second direction, the snap-in connection between the lower plastic member and the upper cover can be effected through the snap-fit, and the connection stability between the lower plastic member and the upper cover can be improved.

[0034] In an implementation, the snap-fit includes an extension portion and a protrusion. The extension portion has one end connected to the first surface and extends in a direction away from the second surface. The protrusion is disposed at another end of the extension portion away from the body and protrudes from a surface of the extension portion positioned facing away from the terminal boss. With this snap-fit structure, the protrusion and the terminal boss are positioned facing away from each other in the second direction, allowing for easy snap-in connection between the snap-fit and the upper cover.

[0035] In an implementation, the lower plastic member further includes a slider portion. The slider portion is disposed on the first surface and located at one side of the extension portion positioned facing away from the terminal boss. The slider portion is configured to allow a slider to slide on the first surface. By providing the slider portion, it is possible to avoid difficulty in demolding or damage to the snap-fit during demolding.

[0036] In an implementation, the lower plastic member further includes an annular push portion. The annular push portion is disposed on the first surface and surrounds the terminal boss. The annular push portion is connected to or spaced apart from the slider portion. In this way, the force can be uniformly applied to the complex terminal boss, thereby ensuring that the terminal boss can be better demolded and avoiding damage.

[0037] In an implementation, the lower plastic member further includes multiple first push-point portions. The multiple first push-point portions are disposed on the first surface and spaced apart from one another. By arranging the multiple first push-point portions spaced apart from one another on the first surface, the force can be uniformly applied to the first surface, thereby facilitating demolding, and preventing the lower plastic member from being damaged due to local uneven stress.

[0038] In an implementation, a bottom wall of the explosion-proof boss recessed from the first surface is a third surface. The third surface defines a vent hole. Each of the first surface and the third surface is provided with multiple second push-point portions spaced apart from one another. By arranging the multiple second push-point portions spaced apart from one another on each of the first surface and the third surface, the force can be uniformly applied to each of the first surface and the third surface, thereby facilitating demolding, and preventing the lower plastic member from being damaged due to local uneven stress.

[0039] In an implementation, a midline extending in a length direction of the first surface is a first midline. A midline extending in the width direction of the first surface is a second midline. The multiple first push-point portions are symmetrical about each of the first midline and the second midline, and/or the multiple second push-point portions are symmetrical about each of the first midline and the second midline. In this way, the multiple first push-point portions can apply force to the lower plastic member more uniformly, and the lower plastic member is not easily deformed during demolding. Similarly, the multiple second push-point portions can apply force to the lower plastic member more uniformly, and the lower plastic member is not easily deformed during demolding.

[0040] In an implementation, the explosion-proof boss includes a first boss and two second bosses. The first boss is disposed at a middle part of the first surface in the width direction of the first surface. Two ends of the first boss in the width direction of the first surface are connected to the two second bosses, respectively. A depth of the first boss recessed relative to the first surface is less than a depth of each second boss recessed relative to the first surface. A bottom wall of the first boss is the third surface. A bottom wall of the second boss recessed from the first surface is a fourth surface. The fourth surface defines multiple air-permeable holes arranged at intervals. The fourth surface is provided with multiple third push-point portions located in the intervals between the multiple air-permeable holes. With this structure of the explosion-proof boss provided, the first boss having a lower protruding height relative to the second surface is provided between the two second bosses. When mounted to the energy storage apparatus, a certain accommodating space can be defined at the first boss between the two second bosses, which is convenient to accommodate partial structures such as cell assemblies, thereby fully utilizing space and reducing the size of the energy storage apparatus. By arranging the multiple third push-point portions at intervals on the fourth surface, the force can be uniformly applied to the fourth surface, thereby facilitating demolding, and preventing the lower plastic member from being damaged due to local uneven stress.

[0041] In an implementation, the lower plastic member further includes a first recessed platform and a second recessed platform. The first recessed platform and the second recessed platform are respectively disposed at two ends of the body in the length direction of the body, and are recessed from the first surface and protrude from the second surface. A bottom wall of the first recessed platform recessed from the first surface is a fifth surface. A bottom wall of the second recessed platform recessed from the first surface is a sixth surface. Each of the fifth surface and the sixth surface defines multiple leakage holes arranged at intervals. Each of the fifth surface and the sixth surface is provided with multiple third push-point portions located in the intervals between the multiple leakage holes. By providing the first recessed platform and the second recessed platform, a top surface of each of the first recessed platform and the second recessed platform protruding from the second surface can be configured to be in contact with the cell assembly, so that there is a certain space between the body and the cell assembly, thereby facilitating accommodation of structures such as a tab. Therefore, the space utilization rate of the energy storage apparatus is increased and the size of the energy storage apparatus is reduced. By arranging the multiple third push-point portions at intervals on each of the fifth surface and the sixth surface, the force can be uniformly applied to each of the fifth surface and the sixth surface, thereby facilitating demolding, and preventing the lower plastic member from being damaged due to local uneven stress.

[0042] In an implementation, each of the multiple first push-point portions is crater-shaped. A middle part of each of the multiple first push-point portions is recessed relative to the first surface. A peripheral edge of each of the multiple first push-point portions protrudes relative to the first surface. By providing the first push-point portion having this shape, demolding can be performed at a higher temperature than room temperature, thereby improving production efficiency.

[0043] In an implementation, the lower plastic member further includes a gate portion. The gate portion is located at one side of the second surface positioned facing away from the first surface. The gate portion is residually formed during injection molding of the lower plastic member. By forming the gate portion at the side of the second surface positioned facing away from the first surface, the first mold is configured for demolding by the push rod, and the second mold is configured for pouring of the plastic liquid, so that the function of the first mold is independent from the function of the second mold. Therefore, misoperation, which is easily caused by performing the pouring of the plastic liquid and the demolding by the push rod on the same mold at the same time, can be avoided, and then adverse effects caused by the misoperation can be avoided.

[0044] In an implementation, the lower plastic member further includes a first recessed platform, and a second recessed platform. The explosion-proof boss is disposed at a middle part of the body

in the length direction of the body. The first recessed platform and the second recessed platform are respectively disposed at two ends of the body in the length direction of the body. Each of the explosion-proof boss, the first recessed platform, and the second recessed platform is recessed from the first surface and protrudes from the second surface. A top surface of each of the explosion-proof boss, the first recessed platform, and the second recessed platform, protruding from the second surface, is provided with the gate portion. By providing the gate portion, the injection molding process can be conveniently carried out, interference with the demolding operation can be avoided, and the product yield can be ensured.

[0045] In a second aspect, an end cover assembly is further provided in the present disclosure. The end cover assembly includes an upper cover and a lower plastic member. The lower plastic member includes a body, a suction portion, and an explosion-proof boss. The body has a first surface and a second surface positioned facing away from the first surface. The first surface of the lower plastic member is positioned facing towards the upper cover. The suction portion is disposed on the first surface. The suction portion includes multiple first ribs and multiple second ribs. The multiple first ribs are parallel to and spaced apart from one another. The multiple second ribs are parallel to and spaced apart from one another. The multiple first ribs and the multiple second ribs are disposed in a crisscross pattern. The multiple first ribs and the multiple second ribs cooperatively define multiple recesses. Each of the multiple recesses is recessed from the first surface towards the second surface. The explosion-proof boss is disposed at a middle part of the body in a length direction of the body. The explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface. The explosion-proof boss is aligned with an explosion-proof valve disposed on the upper cover. A midline extending in a width direction of the first surface is corresponding to the explosion-proof boss. There is a spacing distance between the suction portion and the explosion-proof boss. An end surface of the multiple first ribs positioned facing towards the upper cover is flush with the first surface. An end surface of the multiple second ribs positioned facing towards the upper cover is flush with the first surface. An orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm. An orthographic projection of each of the multiple recesses on the first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm. A depth of each of the multiple recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm. The suction portion is implemented as multiple suction portions. The multiple suction portions are symmetrical about the midline extending in the width direction of the first surface. The multiple suction portions are arranged symmetrically about a second midline.

[0046] In a third aspect, an energy storage apparatus is further provided in the present disclosure. The energy storage apparatus includes a housing, a cell assembly, and the end cover assembly in the second aspect. The cell assembly is disposed in the housing. The end cover assembly is connected to the housing.

[0047] In a fourth aspect, a power consuming device is further provided in the present disclosure. The power consuming device includes a power consuming apparatus and an energy storage apparatus. The energy storage apparatus is configured to supply power to the power consuming apparatus. The energy storage apparatus includes a housing, a cell assembly, and an end cover assembly. The cell assembly is disposed in the housing. The end cover assembly is connected to the housing. The end cover assembly includes an upper cover and a lower plastic member. The lower plastic member includes a body, a suction portion, and an explosion-proof boss. The body has a first surface and a second surface positioned facing away from the first surface. The first surface of the lower plastic member is positioned facing towards the upper cover. The suction portion is disposed on the first surface. The suction portion includes multiple first ribs and multiple second ribs. The multiple first ribs are parallel to and spaced apart from one another. The multiple second ribs are parallel to and spaced apart from one another. The multiple first ribs and the multiple second ribs are disposed in a crisscross pattern. The multiple first ribs and the multiple second ribs

cooperatively define multiple recesses. Each of the multiple recesses is recessed from the first surface towards the second surface. The explosion-proof boss is disposed at a middle part of the body in a length direction of the body. The explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface. The explosion-proof boss is aligned with an explosion-proof valve disposed on the upper cover. A midline extending in a width direction of the first surface is corresponding to the explosion-proof boss. There is a spacing distance between the suction portion and the explosion-proof boss. An end surface of the multiple first ribs positioned facing towards the upper cover is flush with the first surface. An end surface of the multiple second ribs positioned facing towards the upper cover is flush with the first surface. An orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm. An orthographic projection of each of the multiple recesses on the first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm. A depth of each of the multiple recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm. The suction portion is implemented as multiple suction portions. The multiple suction portions are symmetrical about the midline extending in the width direction of the first surface. The multiple suction portions are arranged symmetrically about a second midline.

[0048] In the lower plastic member provided in the present disclosure, the suction portion is provided, and the suction portion is disposed on the first surface and has the recesses recessed from the first surface towards the second surface. During demolding of the mold, the recesses form an effect like the suction cup, so that the suction force between the lower plastic member and the first mold can be increased. The pulling force of the second mold on the lower plastic member is insufficient to separate the lower plastic member from the first mold, thereby avoiding damage to the lower plastic member and improving product yield. The multiple first ribs and the multiple second ribs are disposed to define the multiple recesses. Compared to the single recess with the relatively large size, the multiple dense recesses with relatively small size can have a more stable effect on increasing the suction force with the mold, and have a better effect on avoiding the damage to the lower plastic member and improving the product yield.

[0049] As illustrated in FIG. 1, an energy storage apparatus **1000** provided in an embodiment of the present disclosure is applied to an energy storage system. The energy storage system includes an energy storage apparatus **1000**, an electric energy conversion apparatus (photovoltaic panel **2000**), a wind energy conversion apparatus (wind turbine **3000**), a power grid **4000**, etc. The energy storage apparatus **1000** may serve as an energy storage cabinet, which may be mounted outdoors. Specifically, the photovoltaic panel **2000** can convert solar energy into electric energy during the electricity-price valley period. The energy storage apparatus **1000** is configured to store the electric energy and provide the electric energy to the power grid **4000** during a power-consumption peak period, or supply electricity when there is an outage/blackout in the power grid **4000**. The wind energy conversion apparatus (wind turbine **3000**) can convert wind energy into electric energy. The energy storage apparatus **1000** is configured to store the electric energy and provide the electric energy to the power grid during the power-consumption peak period, or supply electricity when there is an outage/blackout in the power grid. The electric energy may be transmitted using a high voltage cable.

[0050] There may be multiple energy storage apparatuses **1000**, which are connected in series or in parallel. The multiple energy storage apparatuses **1000** are supported and electrically connected by means of a separating plate (not illustrated in the accompanying drawings). In this embodiment, the term “multiple” refers to two or more than two. An energy storage box may be disposed outside the energy storage apparatus **1000**, and used to accommodate the energy storage apparatus **1000**.

[0051] It may be understood that, the energy storage apparatus **1000** may include, but is not limited to a cell, a battery module, a battery pack, a battery system, etc. The actual application form of the energy storage apparatus **1000** provided in embodiments of the present disclosure may be, but is not limited to, the above-listed products, or may be other application forms. Embodiments of the

present disclosure do not impose strict restrictions on the application form of the energy storage apparatus **1000**.

[0052] As illustrated in FIG. **1** and FIG. **10**, a power consuming device **5000** is provided in an embodiment of the present disclosure. The power consuming device **5000** includes a power consuming apparatus **1100** and the energy storage apparatus **1000** in embodiments of the present disclosure. The energy storage apparatus **1000** is configured to supply power to the power consuming apparatus **1100**. The power consuming apparatus **1100** may be a power-generation-side power consuming apparatus, a grid-side power consuming apparatus, a base-station-side power consuming apparatus, a user-side power consuming apparatus, etc. The power consuming apparatus **1100** may specifically be various kinds of power consuming loads, which is not limited herein.

[0053] As illustrated in FIG. **1**, FIG. **2**, and FIG. **9**, an energy storage apparatus **1000** is further provided in embodiments of the present disclosure. The energy storage apparatus **1000** includes a housing **200**, a cell assembly **300**, and an end cover assembly **100** in any embodiment of the present disclosure. The cell assembly **300** is disposed in the housing **200** and the end cover assembly **100** is connected to the housing **200**.

[0054] The housing **200** may be a square housing, and may be made of aluminum or steel, which is not limited. The housing **200** defines an accommodating space. The cell assembly **300** is accommodated in the accommodating space. The accommodating space has an opening(s) at one end or two ends of the accommodating space. The end cover assembly **100** is disposed at the opening. The end cover assembly **100** is connected and fixed to the housing **200**, and close the opening. The cell assembly **300** may specifically include one or more cells connected to tab(s). The end cover assembly **100** may include an upper cover **20**, a lower plastic member **10**, a terminal (not shown), an explosion-proof valve **23**, and the like. The lower plastic member **10** is disposed at one side of the upper cover **20** facing the cell assembly **300**. The terminal passes through the lower plastic member **10** and the upper cover **20**. The terminal is insulated from the upper cover **20**. The terminal is connected to the tab. An external power source is connected to the terminal to charge the battery cell through the tab. The electric energy of the battery cell can also be transmitted to an external power consuming load through the tab and the terminal. The explosion-proof valve is disposed on the upper cover **20**. When a large amount of gas is generated due to abnormal heat of the cell assembly **300**, the explosion-proof valve can be exploded to release the gas, so as to avoid the explosion of the energy storage apparatus.

[0055] As illustrated in FIG. **2** and FIG. **9**, an end cover assembly **100** is further provided in embodiments of the present disclosure. The end cover assembly **100** includes an upper cover **20** and the lower plastic member **10** in any embodiment of the present disclosure. A first surface **111** of the lower plastic member **10** faces the upper cover **20**. The end cover assembly **100** can be referred to above, and is not limited herein.

[0056] The lower plastic member **10** in embodiments of the present disclosure will be introduced in detail below.

[0057] As illustrated in FIG. **2** and FIG. **3**, the lower plastic member **10** is provided in embodiments of the present disclosure. The lower plastic member **10** includes a body **11** and a suction portion **12**. The lower plastic member **10** is made by an injection molding process. Various structures of the lower plastic member **10** described below are integrated structures formed by injection molding.

[0058] The body **11** has a first surface **111** and a second surface **112** positioned facing away from the first surface **111**. The first surface **111** is configured to be positioned at one side of the body **11** facing towards an upper cover. The second surface **112** is configured to be positioned at one side of the body **11** facing towards a cell assembly. The body **11** substantially has a sheet-like structure. The first surface **111** and second surface **112** are substantially flat or slightly curved, which are not limited herein. The first surface **111** and second surface **112** each are substantially rectangular.

[0059] A coordinate system is established for convenience of explanation. A first direction X is a

length direction of the body **11**, and is also a length direction of each of the first surface **111**, the second surface **112**, and the lower plastic member **10**. A second direction Y is a width direction of the body **11**, and is also a width direction of each of the first surface **111**, the second surface **112**, and the lower plastic member **10**. A third direction Z is a thickness direction of the body **11**, that is, a direction in which the first surface **111** and the second surface **112** that are positioned facing away from each other, and a thickness direction of the lower plastic member **10**.

[0060] As illustrated in FIG. 2 and FIG. 7, the suction portion **12** is disposed on the first surface **111**. The suction portion **12** has recesses **121** each recessed from the first surface **111** towards the second surface **112**. Each recess **121** is slightly recessed from the first surface **111** and does not extend through the body to the second surface **112**. The depth of each recess **121** is negligible with respect to the thickness of the body **11**. Therefore, the presence of the recesses **121** does not affect the strength of the lower plastic member **10**.

[0061] As illustrated in FIG. 3 and FIG. 4, the lower plastic member **10** is formed by injection molding after the first mold **21** (fixed mold) and the second mold **22** (movable mold) are closed, and a plane where the first mold **21** and the second mold **22** are butted against each other is a mold-mating plane **201**. The mold-mating plane **201** is shown by a dotted line in FIG. 4. In embodiments of the present disclosure, the first mold **21** and the second mold **22** are butted in the third direction Z (i.e., the thickness direction of the lower plastic member **10**), and the mold-mating plane **201** coincides with the first surface **111**. After injection molding is completed, a demolding direction is also the third direction Z.

[0062] The first mold **21** has a first molding surface **211**. The first molding surface **211** is flat and extends to an edge of the first mold **21**. The first mold **21** may define a recessed first molding groove **212** on the first molding surface **211**. The first molding groove **212** is corresponding to a structure of the lower plastic member **10** protruding from the first surface **111**. According to the quantity and different shapes of the structures of the lower plastic member **10** protruding from the first surface **111**, the quantity and shape of the first molding grooves **212** can be designed adaptively, which are not specifically limited.

[0063] The second mold **22** has a second molding surface **221**. The second molding surface **221** is flat, and is at a certain distance from an edge of the first mold **21** in a middle region of the first mold **21**. The second mold **22** may define a recessed second molding groove **222** on the second molding surface **221**. The second molding groove **222** is corresponding to a structure of the lower plastic member **10** protruding from the second surface **112**. According to the quantity and different shapes of the structures of the lower plastic member **10** protruding from the second surface **112**, the quantity and shape of the second molding grooves **222** can be designed adaptively, which are not specifically limited.

[0064] A butting portion **223** protruding from the second molding surface **221** is further positioned at a peripheral edge of the second mold **22**. The second molding surface **221** can extend to and be connected to a side surface of the butting portion **223**. A top surface of the butting portion **223** protruding from the first surface **111** is a butting surface **224**. The butting surface **224** is configured to be in close contact with the first molding surface **211** of the first mold **21**.

[0065] During injection molding, the first mold **21** and the second mold **22** are butted, so that the butting surface **224** is in close contact with the first molding surface **211**. At this time, there is a gap between the first molding surface **211** and the second molding surface **221** to define a cavity. The first molding groove **212** and the second molding groove **222** both are in communication with the cavity to jointly form a mold cavity. Molten plastic liquid is poured into the mold cavity, and is cooled after the mold cavity is filled with the molten plastic liquid, and the lower plastic member **10** can be formed after the plastic liquid is cooled and solidified. The first molding surface **211** is configured to form the first surface **111** of the lower plastic member **10**. The second molding surface **221** is configured to form the second surface **112** of the lower plastic member **10**. The first molding groove **212** is configured to form the structure of the lower plastic member **10** protruding

from the first surface **111**. The second molding groove **222** is configured to form the structure of the lower plastic member **10** protruding from the second surface **112**.

[0066] During demolding, the second mold **22** is moved away from the first mold **21** to separate the second mold **22** from the lower plastic member **10**, and then the lower plastic member **10** is pushed away from the first mold **21** by a push rod (not shown) to separate the lower plastic member **10** from the first mold **21**. During separation of the second mold **22** from the lower plastic member **10**, the lower plastic member **10** may be pulled by the second mold **22** to a certain extent, so that the lower plastic member **10** to be separated from the first mold **21**, and further the lower plastic member **10** may be locally deformed and damaged, resulting in low product yield.

[0067] In embodiments of the present disclosure, during injection molding, it is not necessary to wait for the plastic liquid to cool to room temperature before demolding, but rather, it is only necessary for the plastic liquid to solidify and mold and to be demolded at a higher temperature, so that the production efficiency can be improved. However, the plastic that is not completely cooled to room temperature may leave some marks to form specific shapes when the plastic is subjected to external force. Thanks to the optimization of injection molding process, these marks do not affect the functional realization of the lower plastic member **10**, and the production efficiency can be significantly improved.

[0068] In embodiments of the present disclosure, the suction portion **12** is disposed on the first surface **111**. The suction portion **12** has the recesses **121**. Each recess **121** is also formed by a protruding structure (not shown) protruding from the first molding surface **211** on the first mold **21**. In other words, before demolding, the protruding structure of the first mold **21** is embedded in the lower plastic member **10**.

[0069] Since the suction portion **12** is a structure left after injection molding and has the recesses **121**, the suction portion **12** differs significantly in shape compared to other regions of the first surface **111**, so that the suction portion **12** can be easily distinguished from the other regions of the first surface **111**. During the assembly of the end cover assembly, the suction portion **12** can be serve as a recognition feature for machine vision system of automated production lines, thereby facilitating quick positioning of the lower plastic member **10** and determining whether the position of the lower plastic member **10** is accurate.

[0070] In addition, after the lower plastic member **10** is assembled to form the energy storage apparatus **1000**, when the electrolyte is filled into the energy storage apparatus **1000**, a small amount of electrolyte spreads along a gap between the upper cover and the lower plastic member **10** and enters the above-mentioned recesses **121**. Due to the surface tension of the electrolyte, the lower plastic member **10** and the upper cover can be attached together. At this time, the region where the recesses **121** are located forms a locally enclosed space and filled with electrolyte. If the lower plastic member **10** has a falling tendency (that is, when the lower plastic member **10** is about to be separated from the upper cover), the recesses **121** form a negative pressure region to resist the falling tendency of the lower plastic member **10**. Since the recesses **121** are filled with electrolyte, gas can be prevented from entering and disrupting the negative pressure region. Therefore, the lower plastic member **10** is firmly attached onto a lower surface of the upper cover (that is, a surface of the upper cover positioned facing towards the lower plastic member **10**), the structural strength of the end cover assembly is improved, and a Myra film welded to a sidewall of the lower plastic member **10** is prevented from being deformed.

[0071] Correspondingly, the lower surface of the upper cover of the end cover assembly (that is, a surface of the upper cover positioned facing towards the lower plastic member **10**) is relatively smooth, with a surface roughness less than or equal to Ra3.2. Therefore, the first surface **111** is more tightly attached to the lower surface of the upper cover, and the negative pressure region is not easily damaged by air leakage.

[0072] Therefore, in the lower plastic member **10** provided in embodiments of the present disclosure, the suction portion **12** is provided, and the suction portion **12** is disposed on the first

surface **111** and has the recesses **121** recessed from the first surface **111** towards the second surface **112**. During demolding of the mold, the recesses **121** form an effect like a suction cup, so that the suction force between the lower plastic member **10** and the first mold **21** can be increased. The pulling force of the second mold **22** on the lower plastic member **10** is insufficient to separate the lower plastic member **10** from the first mold **21**, thereby avoiding damage to the lower plastic member **10** and improving product yield.

[0073] In an embodiment, as illustrated in FIG. 2 and FIG. 7, the suction portion **12** includes multiple first ribs **122** and multiple second ribs **123**. The multiple first ribs **122** are parallel to and spaced apart from one another. The multiple second ribs **123** are parallel to and spaced apart from one another. The multiple first ribs **122** and the multiple second ribs **123** are disposed in a crisscross pattern. The multiple first ribs **122** and the multiple second ribs **123** cooperatively define multiple recesses **121**.

[0074] Optionally, each first rib **122** extends in the first direction X. Each second rib **123** extends in the second direction Y. The multiple first ribs **122** are arranged in the second direction Y at equal intervals. The multiple second ribs **123** are arranged in the first direction X at equal intervals.

[0075] Optionally, the multiple first ribs **122** may be arranged at non-equal intervals, and the multiple second ribs **123** may be arranged at non-equal intervals, all of which are not limited.

[0076] An angle defined by the intersection of each first rib **122** and each second rib **123** may be any angle other than 0°, such as 30°, 45°, 60°, 90°, etc., which is not limited.

[0077] Each of the multiple first ribs **122** and the multiple second ribs **123** may be flush with the first surface **111**, or slightly protrude from the first surface **111** or be slightly recessed relative to the first surface **111**, which is not limited. The thicknesses of the multiple first ribs **122** and the multiple second ribs **123** may or may not be equal, which are not limited.

[0078] By providing the multiple first ribs **122** and the multiple second ribs **123** to define the multiple recesses **121**, compared to a single recess **121** with a relatively large size, the multiple dense recesses **121** with relatively small size can have a more stable effect on increasing the suction force with the first mold **21** (as illustrated in FIG. 4 and FIG. 5), and have a better effect on avoiding damage to the lower plastic member **10** and improving the product yield.

[0079] Optionally, an end portion of each of the multiple first ribs **122** and the multiple second ribs **123**, configured to be positioned facing towards the upper cover, has a pointed-prism shape. Specifically, the thickness of each of the multiple first ribs **122** and the multiple second ribs **123** gradually decreases from a bottom wall of each of the multiple recesses **121** towards an opening of the same.

[0080] This arrangement ensures that the size of the recess **121** at the opening is larger than the size of the recess **121** at a deeper position in a depth direction of the recess **121**. In other words, the recess **121** has an outward-flared structure. The outward-flared structure of the recess **121** is related to the protruding structure provided on the first mold **21**. In other words, the size of the top of the protruding structure (corresponding to the bottom wall of the recess **121**) is smaller, and the size of the base of the protruding structure (corresponding to the opening of the recess **121**) is larger. In this way, during demolding of the lower plastic member **10** from the first mold **21**, once the lower plastic member **10** is slightly separated from the first mold **21**, a gap can be defined between the protruding structure corresponding to the recess **121** on the lower plastic member **10** and the sidewall of the recess **121**. Therefore, air can enter the recess **121**, the first rib **122** and the second rib **123** can be prevented from always being in close contact with the protruding structure of the mold, and the relatively large negative pressure generated between the protruding structure of the first mold **21** and the recess **121** can be reduced, thereby facilitating demolding.

[0081] Optionally, as illustrated in FIG. 2 and FIG. 7, an orthographic projection of the suction portion **12** on the first surface **111** is circular and has a diameter ranging from 5 mm to 10 mm. The diameter of the suction portion **12** may be specifically 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, 10 mm, etc., which is not limited. The circular suction portion **12** provided is simple in shape, so that the

protruding structure, on the first molding surface **211** of the first mold **21** (as illustrated in FIG. 4 and FIG. 5), for forming the suction portion **12** is relatively easy to manufactured in shape.

[0082] Optionally, the orthographic projection of the suction portion **12** on the first surface **111** may have another shape, which is not limited.

[0083] Optionally, as illustrated in FIG. 2 and FIG. 7, an orthographic projection of each of the multiple recesses **121** on the first surface **111** is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm. The shape of the orthographic projection of the recess **121** on the first surface **111** refers to the contour shape of the peripheral sidewall of the recess **121**. When a distance between two adjacent first ribs **122** is equal to a distance between two adjacent second ribs **123**, the shape of the recess **121** is square. When the distance between the two adjacent first ribs **122** is unequal to the distance between the two adjacent second ribs **123**, the shape of the recess **121** is rectangular. The length of the rectangular recess **121** may be 0.3 mm, 0.4 mm, 0.45 mm, 0.5 mm, 0.6 mm, etc., which is not limited. The width of the rectangular recess **121** may be 0.3 mm, 0.4 mm, 0.45 mm, 0.5 mm, 0.6 mm, etc., which is not limited.

[0084] It may be understood that, near the peripheral edge of the suction portion **12**, the shape of the recess **121** is a partial rectangular shape, rather than a complete rectangular shape. This is because the edge of the suction portion **12** intersects with the first ribs **122** and the second ribs **123**, thereby dividing the shape of the rectangular shape.

[0085] The orthographic projection on the first surface **111** is set to be in the shape of the recess **121**, and each of the length and width of the recess **121** is set to range from 0.3 mm to 0.6 mm. Therefore, the recess **121** is simple in shape and reasonable in size, which is convenient for injection molding and also convenient for manufacturing of the mold.

[0086] Optionally, the orthographic projection of the recess **121** on the first surface **111** may have another shape other than a rectangular shape, and the dimensions of the length and width of the recess **121** may have other values, which are not limited.

[0087] Optionally, as illustrated in FIG. 2 and FIG. 7, a depth of each of the multiple recesses **121** recessed from the first surface **111** ranges from 0.2 mm to 0.3 mm. The depth of the recess **121** may be 0.2 mm, 0.21 mm, 0.22 mm, 0.23 mm, 0.24 mm, 0.25 mm, 0.26 mm, 0.27 mm, 0.28 mm, 0.29 mm, 0.3 mm, etc., which is not limited. Due to the relatively thin thickness of the body **11** of the lower plastic member **10**, by setting the recess **121** to have a relatively shallow depth, the suction force between the lower plastic member **10** and the first mold **21** is increased, and the impact of the recess **121** on the strength of the lower plastic member **10**.

[0088] Optionally, the depth of the recess **121** recessed from the first surface **111** may be other values, which is not limited.

[0089] In a specific embodiment, as illustrated in FIG. 2 and FIG. 7, the diameter of the suction portion **12** is 7 mm. The rectangular orthographic projection of the recesses **121** on the first surface **111** has the length of 0.45 mm and the width of 0.45 mm. The depth of the recess **121** recessed from the first surface **111** is 0.25 mm.

[0090] By setting each dimension of the above suction portion **12**, the suction force between the lower plastic member **10** and the mold can be increased, and the damage to the lower plastic member **10** caused by the bending and deformation of the lower plastic member **10** can be avoided. In addition, the structure is simple, the manufacturing is convenient, and the cost is low. In an embodiment, as illustrated in FIG. 2 and FIG. 7, there are multiple suction portions **12**. The multiple suction portions **12** are symmetrical about a midline extending in the width direction of the first surface **111**.

[0091] The first surface **111** has a first midline A1 extending in the length direction of the lower plastic member **10** and a second midline A2 extending in the width direction of the lower plastic member **10**. The multiple suction portions **12** are arranged symmetrically about the second midline A2. The multiple suction portions **12** symmetrically arranged feature a simple layout, thereby facilitating the mold manufacturing and also facilitating visual recognition during the subsequent

assembly of the end cover assembly.

[0092] Optionally, as illustrated in FIG. 2 and FIG. 7, at least two of the multiple suction portions **12** are disposed on a same side of the midline (that is, the second midline A2) extending in the width direction of the first surface **111**, and have different distances from the midline (that is, the second midline A2) extending in the width direction of the first surface **111**.

[0093] Exemplarily, as illustrated in FIG. 7, there are four suction portions **12**. Two suction portions **12** are provided in the vicinity of each of two side edges of the body **11** in the width direction of the body **11**. The four suction portions **12** are asymmetrical about the first midline A1, but are symmetrical about the second midline A2. Two suction portions **12** located at the same side of the second midline A2 are disposed in a staggered manner in the second direction Y, that is, there is a spacing distance L between the two suction portions **12** in the first direction X. Optionally, the spacing distance L may range from 3 mm to 3.5 mm, and specifically may be 3 mm, 3.1 mm, 3.2 mm, 3.25 mm, 3.3 mm, 3.4 mm, 3.5 mm, etc., which is not limited.

[0094] This arrangement of the multiple suction portions **12** provides a foolproof function for machine vision recognition in automated production lines during the assembly of the end cover assembly, thereby preventing the lower plastic member **10** from being placed incorrectly in a reversed left-right orientation.

[0095] In an embodiment, as illustrated in FIG. 2 and FIG. 7, the first surface **111** is provided with a code portion **13**. The code portion **13** has a preset shape. The preset shape includes at least one of a number, a letter, a bar code, or a two-dimensional code.

[0096] The code portion **13**, similar to the suction portion **12** described above, is also a structure left after injection molding. The shape of the code portion **13** may be formed by protruding or being recessed from the first surface **111**, and a corresponding shape can be set on the first molding surface **211** of the first mold **21** (as illustrated in FIG. 4 and FIG. 5) in a manner similar to movable type printing, so that the code portion **13** is formed on the lower plastic member **10** after injection molding. The preset shape of the code portion **13** can be set as necessary. For example, the shape of the code portion **13** may be a number, a letter, or the like that can be visually recognized directly with the naked eyes, or may be a bar code, a two-dimensional code, or the like that can be recognized by a terminal equipment scanning the code. By providing the code portion **13**, which can be recognized with the naked eye or recognized by scanning the code, the product category, production date, manufacturer name, and other characteristics of the lower plastic member **10** can be obtained, thereby facilitating the traceability of the lower plastic member **10**, and the control of the production process and the traceability of quality issues.

[0097] Optionally, referring to FIGS. 2 and 7, there are multiple code portions **13** arranged side by side.

[0098] Optionally, each of the multiple code portions **13** may be substantially rectangular, the length direction thereof being the first direction X, and the width direction thereof being the second direction Y. The multiple code portions **13** may be sequentially arranged in the second direction Y. Each code portion **13** may have a preset shape, and preset shapes of the multiple code portions **13** may be different from each other. In other words, each code portion **13** can recognize one kind of information, and kinds of information recognized by the multiple code portions **13** are different from each other. The length and width of the multiple code portions **13** are not limited, and may be provided as necessary. For example, as illustrated in FIG. 7, the widths (that is, the dimensions in the second direction Y) of the multiple code portions **13** are substantially equal, and the length (that is, the dimension in the first direction X) of each of some of the multiple code portions **13** is longer than the length of each of the rest of the multiple code portions **13**. The multiple code portions **13** having the same length may be provided with both ends aligned or staggered, which is not limited.

[0099] During injection molding, multiple replaceable code blocks are provided on the first mold **21**. Each code block is replaceable. According to the preset shape corresponding to the information required to be recognized by each lower plastic member **10**, a corresponding shape is formed on a

corresponding replaceable code block. For example, for the production date information, since the date changes daily, there are multiple replaceable code blocks, and different replaceable code blocks can be replaced according to different dates, so that the code portion **13** of the lower plastic member **10** produced on different dates presents different dates. For other kinds of information, multiple replaceable code blocks may be provided to present different kinds of information. [0100] By arranging the multiple code portions **13** side by side, different pieces of information can be displayed or recognized in one position, which is convenient for tracing various kinds of information.

[0101] Optionally, the first surface **111** of the body **11** may be provided with multiple code regions. Each code region is provided with at least one code portion **13**. As illustrated in FIG. 7, one code region is provided on each of the left and right sides of the second midline A2 (that is, two opposite sides in the first direction X). Each code region is provided with multiple code portions **13**. In structure, the two code regions is symmetrical about the second midline A2. The multiple code regions may be configured to display different kinds of information or the same kind of information, which is not limited.

[0102] Optionally, as illustrated in FIG. 2 and FIG. 7, any adjacent two of the multiple code portions **13** are spaced apart from each other. Taking the multiple code portions **13** illustrated in FIG. 7 as an example, the multiple code portions **13** are sequentially arranged in the second direction Y, and there is a gap between any two adjacent code portions **13**.

[0103] During demolding, after the second mold **22** is separated from the first mold **21** and the lower plastic member **10**, the lower plastic member **10** needs to be separated from the first mold **21**. The multiple code portions **13** correspond to multiple replaceable code blocks on the first mold **21**. By providing a gap between any two adjacent code portions **13**, or in other words, between any two adjacent movable code portions, air can enter the gap during demolding. Therefore, the local negative pressure between the replaceable code blocks and the lower plastic member **10** is avoided, the demolding resistance between the lower plastic member **10** and the first mold **21** is reduced, and the production efficiency is improved.

[0104] In an embodiment, as illustrated in FIG. 2 and FIG. 3, the lower plastic member **10** further includes a terminal boss **14**. The terminal boss **14** is connected to the body **11** and projects from the first surface **111**. A snap-fit **15** is provided at each of two sides of the terminal boss **14** in the width direction of the first surface **111**.

[0105] Specifically, the terminal boss **14** includes a first terminal boss **141** and a second terminal boss **144**. Terminals of the end cover assembly include a first terminal (not shown) and a second terminal (not shown), with opposite polarities. The first terminal boss **141** protrudes relative to the first surface **111** and is recessed relative to the second surface **112**, to define a first terminal-groove **143**. The first terminal boss **141** defines a first terminal-hole **142** extending through the first terminal boss **141** in the third direction Z. The first terminal-hole **142** allows the first terminal to pass through. The first terminal-groove **143** is configured to accommodate a base structure (not shown) of the first terminal. The second terminal boss **144** protrudes relative to the first surface **111** and is recessed relative to the second surface **112**, to define a second terminal-groove **146**. The second terminal boss **144** defines a second terminal-hole **145** extending through the second terminal boss **144** in the third direction Z. The second terminal-hole **145** allows for the second terminal to pass through. The second pole recess **146** is configured to accommodate a base structure (not shown) of the second terminal. One snap-fit **15** is provided at each of two sides of the first terminal boss **141** in the second direction Y, and one snap-fit **15** is also provided at each of two sides of the second terminal boss **144** in the second direction Y, so that the lower plastic member **10** includes four snap-fits **15** in total, and the four snap-fits **15** are symmetrical about the first midline A1 and the second midline A2. As described in the above embodiments, the terminal boss, the snap-fit **15**, and the like, are also formed by injection molding together with the body **11**, that is, various structures of the lower plastic member **10** are integrated, which will not be repeatedly

described later.

[0106] When the lower plastic member **10** is assembled with the upper cover, the first surface **111** of the lower plastic member **10** is positioned facing towards the upper cover, and the upper cover may define a groove for accommodating the terminal boss **14**, so that the first surface **111** is in close contact with the upper cover. Additionally, the upper cover may be provided with a snap structure in fit with the snap-fit **15**. When the first surface **111** of the lower plastic member **10** is in close contact with the upper cover, the snap-fit **15** is snap-fitted with and fixed to the snap structure, thereby improving the connection stability between the upper cover and the lower plastic member **10**.

[0107] By providing the snap-fit **15** at each of the two sides of the terminal boss **14** in the second direction Y, the snap-in connection between the lower plastic member **10** and the upper cover can be effected through the snap-fit **15**, and the connection stability between the lower plastic member **10** and the upper cover can be improved.

[0108] In an embodiment, as illustrated in FIG. 2 and FIG. 3, the snap-fit **15** includes an extension portion **151** and a protrusion **152**. The extension portion **151** has one end connected to the first surface **111** and extends in a direction away from the second surface **112**. The protrusion **152** is disposed at one another of the extension portion **151** away from the body **11** and protrudes from a surface of the extension portion **151** positioned facing away from the terminal boss **14**.

[0109] With this structure of the snap-fit **15**, the protrusion **152** and the terminal boss **14** are positioned facing away from each other in the second direction Y, allowing for easy snap-in connection between the snap-fit **15** and the upper cover.

[0110] Optionally, as illustrated in FIG. 2, an avoidance notch **147** may be defined on each of two opposite side surfaces of the terminal boss **14** in the second direction Y. The avoidance notch **147** can accommodate at least a part of the structure of the snap-fit **15**. There is a spacing distance between the snap-fit **15** and an inner wall of the avoidance notch **147**. In this way, the space occupied by the snap-fit **15** can be saved, and the function of the terminal boss **14** is not affected. When the snap-fit **15** is snap-fitted with the upper cover, the protrusion **152** of the snap-fit **15** is squeezed, resulting in resilient deformation of a top end of the snap-fit **15** away from the first surface **111** towards the terminal boss. By designing the spacing distance between the snap-fit **15** and the avoidance notch **147**, interference between the snap-fit **15** and the terminal boss can be avoided.

[0111] Optionally, as illustrated in FIG. 2 and FIG. 7, the lower plastic member **10** further includes a slider portion **16**. The slider portion **16** is disposed on the first surface **111** and located at one side of the extension portion **151** positioned facing away from the terminal boss **14**. The slider portion **16** is configured to allow a slider to slide on the first surface **111**.

[0112] The slider portion **16**, similar to the suction portion **12** and the code portion **13** described above, is also a structure left on the first surface **111** due to the action of the mold on the first surface **111** during injection molding. In order to mold the snap-fit **15**, the first mold **21** is provided with a slider, which is movable in the second direction Y of the lower plastic member **10**. During injection molding, the slider is positioned closer to the terminal boss **14**. When demolding after the injection molding is completed, if the slider is not moved and the lower plastic member **10** is directly separated from the first mold **21**, the interference will occur between the slider and the protrusion **152** of the snap-fit **15**, making demolding difficult. As a result, forced demolding may cause the protrusion **152** to be worn down. To prevent this, during demolding, the slider is moved away from the terminal boss **14** in the second direction Y to avoid the protrusion **152** of the snap-fit **15**. A mark left on the first surface **111** by sliding of the slider from a position during injection molding to a position during demolding is the slider portion **16**.

[0113] The slider portion **16** may be a protruding structure or a recessed structure relative to the first surface **111**. An orthographic projection of the slider portion **16** on the first surface **111** may be rectangular or any other shape, which is not limited.

[0114] By providing the slider portion **16**, it is possible to avoid difficulty in demolding or damage to the snap-fit **15** during demolding.

[0115] In an implementation, as illustrated in FIG. 2 and FIG. 7, the lower plastic member **10** further includes an annular push portion **171**. The annular push portion **171** is disposed on the first surface **111** and surrounds the terminal boss **14**. The annular push portion **171** is connected to or spaced apart from the slider portion **16**.

[0116] As illustrated in FIG. 4 and FIG. 5, the first mold **21** defines a push hole **213** (e.g., an ejector pin hole) extending through the first mold **21** in the thickness direction (that is, the third direction Z) of the lower plastic member **10**. During injection molding, the push hole **213** can be blocked by a plunger and other structures to prevent plastic liquid from entering the push hole **213**. When injection molding is completed, the plunger is pulled out and the push rod (e.g., an ejector pin) is inserted into the push hole **213** to push the lower plastic member **10** away from the first mold **21**. The pressure of the push rod acting on the lower plastic member **10** will leave marks on the lower plastic member **10** after the action of the push rod. The annular push portion **171** in this embodiment is a residual mark after demolding, and various push-point portion in the following description are also residual marks after demolding, all of which are obtained by adopting a relatively high-temperature demolding process.

[0117] Due to the relatively complex structure of the terminal boss **14**, push rods are adopted to surround the periphery of the terminal boss **14** to press against the first surface **111**, to form the annular push portion **171** around the terminal boss. In this way, the force can be uniformly applied to the complex terminal boss **14**, thereby ensuring that the terminal boss **14** can be better demolded and avoiding damage.

[0118] Due to the need for injection molding of the snap-fit **15**, a slider for injection molding needs to be provided at a position corresponding to the slider portion **16**. During demolding, the slider needs to be separated along with the lower plastic member **10**. Therefore, when the push rod acts on the first surface **111**, the push rod cannot be pressed against the slider, but needs to be in contact with or spaced apart from the side surface of the slider. Thus, the remaining annular push portion **171** is connected to or spaced apart from the remaining slider portion **16**.

[0119] In an implementation, referring to FIG. 2, the lower plastic member **10** further includes multiple first push-point portions **172**. The multiple first push-point portions **172** are disposed on the first surface **111** and spaced apart from one another.

[0120] The first push-point portion **172** is also a structure left after injection molding, that is, a mark left on the first surface **111** after the push rod acts on the first surface **111**. The shape of the first push-point portion **172** may be circular. The diameter of the first push-point portion **172** may range from 4.5 mm to 5 mm, specifically 4.5 mm, 4.6 mm, 4.7 mm, 4.8 mm, 4.9 mm, 5 mm, etc. In a specific embodiment, the diameter of the first push-point portion **172** is 4.85 mm. The first push-point portion **172** may have other shapes and sizes, which are not limited.

[0121] By arranging the multiple first push-point portions **172** spaced apart from one another on the first surface **111**, the force can be uniformly applied to the first surface **111**, thereby facilitating demolding, and preventing the lower plastic member **10** from being damaged due to local uneven stress.

[0122] Optionally, as illustrated in FIG. 7 and FIG. 8, each of the multiple first push-point portions **172** is crater-shaped. A middle part of each of the multiple first push-point portions **172** is recessed relative to the first surface **111**. A peripheral edge of each of the multiple first push-point portions **172** protrudes relative to the first surface **111**.

[0123] Due to high-temperature demolding, the lower plastic member **10** is not completely cooled to room temperature, and the lower plastic member **10** is relatively soft although solidified and molded. When the push rod acts on the first surface **111**, the first surface **111** will be slightly deformed under the pressure of the push rod, thereby forming a crater-shaped structure.

[0124] The crater-shape first push-point portion **172** has a recessed depth that is negligible with

respect to the thickness of the body **11**. The height of the crater-shape first push-point portion **172** protruding relative to the first surface **111** is also negligible with respect to the thickness of the body **11**. By providing the first push-point portion **172** having this shape, demolding can be performed at a higher temperature than room temperature, thereby improving production efficiency. [0125] The structure of each subsequent push-point portion is similar to the structure of the first push-point portion **172**, each having a crater shape, which will not be repeated later.

[0126] Optionally, the multiple first push-point portions **172** are symmetrical about each of the first midline A1 and the second midline A2. In this way, the multiple first push-point portions **172** can apply force to the lower plastic member **10** more uniformly, and the lower plastic member **10** is not easily deformed during demolding.

[0127] As illustrated in FIG. 2, FIG. 3, and FIG. 7, the lower plastic member **10** further includes an explosion-proof boss **18**. The explosion-proof boss **18** is disposed at a middle part of the body **11** in a length direction of the body **11**. The explosion-proof boss **18** is recessed relative to the first surface **111** and protrudes relative to the second surface **112**. A bottom wall of the explosion-proof boss **18** recessed from the first surface **111** is a third surface. The third surface **182** defines a vent hole **183** extending through the third surface **182** in the third direction Z. Each of the first surface **111** and the third surface **182** is provided with multiple second push-point portions **173** spaced apart from one another.

[0128] The explosion-proof boss **18** is configured to be aligned with an explosion-proof valve disposed on the upper cover. The explosion-proof boss **18** is recessed from the first surface **111**, to define a gas collecting chamber. The gas generated by the cell assembly can flow to the gas collecting chamber through the vent hole **183**. If the air pressure in the gas collecting chamber is too high, the explosion-proof valve will be triggered to explode and deflate.

[0129] The specific structure of the explosion-proof boss **18** may not be limited. The explosion-proof boss **18** may substantially extend in the second direction Y, and may be symmetrical about each of the first midline A1 and the second midline A2. An intersection of the first midline A1 and the second midline A2 is located at the explosion-proof boss **18**. The vent hole **183** may be circular, square, arc-shaped, or the like, which is not limited. The vent hole **183** includes multiple smaller holes, so that the explosion-proof boss **18** forms a fence-like structure. Therefore, the explosion-proof boss **18** meets the requirements of structural strength, allows ventilation, and prevents a foreign object on the cell assembly from passing through the vent hole **183** to avoid a short circuit between the foreign object and the upper cover.

[0130] The second push-point portion **173** is also a structure left after injection molding, that is, a mark left after the push rod acts on each of the first surface **111** and the third surface **182**. The shape of the second push-point portion **173** may be circular. The diameter of the second push-point portion **173** may range from 3 mm to 4 mm, specifically 3 mm, 3.2 mm, 3.4 mm, 3.6 mm, 3.8 mm, 4 mm, etc. The second push-point portion **173** may have other shapes and sizes, which are not limited.

[0131] The second push-point portions **173** on the first surface **111** may be located on opposite sides of the suction portion **12** in the first direction X. The push-point portions on the third surface **182** may be located in the intervals between the multiple vent holes **183**, or may be located outside the entire set of vent holes **183**, which is not limited.

[0132] By arranging the multiple second push-point portions **173** spaced apart from one another on each of the first surface **111** and the third surface **182**, the force can be uniformly applied to each of the first surface **111** and the third surface **182**, thereby facilitating demolding, and preventing the lower plastic member **10** from being damaged due to local uneven stress.

[0133] Optionally, the multiple second push-point portions **173** are symmetrical about each of the first midline A1 and the second midline A2. In this way, the multiple second push-point portions **173** can apply force to the lower plastic member **10** more uniformly, and the lower plastic member **10** is not easily deformed during demolding.

[0134] In an embodiment, as illustrated in FIG. 2, FIG. 3, and FIG. 7, the explosion-proof boss **18** includes a first boss **181** and two second bosses **184**. The first boss **181** is disposed at a middle part of the first surface **111** in a width direction of the first surface **111**. Two ends of the first boss **181** in the width direction of the first surface **111** are connected to the two second bosses **184**, respectively. A depth of the first boss **181** recessed relative to the first surface **111** is less than a depth of each second boss **184** recessed relative to the first surface **111**. Taking the first surface **111** as a reference, the depth of the first boss **181** recessed is shallower, and the depth of the second boss **184** recessed is deeper. Taking the second surface **112** as a reference, the height of the first boss **181** protruding is lower, and the height of the second boss **184** protruding is higher.

[0135] With this structure of the explosion-proof boss **18** provided, the first boss **181** having a lower protruding height relative to the second surface **112** is provided between the two second bosses **184**. When mounted to the energy storage apparatus, a certain accommodating space can be defined at the first boss **181** between the two second bosses **184**, which is convenient to accommodate partial structures such as cell assemblies, thereby fully utilizing space and reducing the size of the energy storage apparatus.

[0136] A bottom wall of the first boss **181** is the third surface **182**. A bottom wall of the second boss **184** recessed from the first surface **111** is a fourth surface **185**. The fourth surface **185** defines multiple air-permeable holes **186** arranged at intervals. The fourth surface **185** is provided with multiple third push-point portions **174** located in the intervals between the multiple air-permeable holes **186**.

[0137] The air-permeable hole **186** functions similarly to the vent hole **183** described above, and is also used for ventilation. The shape of the vent hole **183** may be circular or other shapes, and the size of the vent hole **183** is not limited.

[0138] The third push-point portion **174** is also a structure left after injection molding, that is, a mark left after the push rod acts on the fourth surface **185**. The shape of the third push-point portion **174** may be circular. The diameter of the third push-point portion **174** may range from 2 mm to 2.5 mm, specifically 2 mm, 2.1 mm, 2.2 mm, 2.3 mm, 2.4 mm, 2.5 mm, etc. In a specific embodiment, the diameter of the third push-point portion **174** is 2.25 mm. The third push-point portion **174** may have other shapes and sizes, which are not limited.

[0139] According to arrangement positions of the multiple air-permeable holes **186**, the multiple third push-point portions **174** may be symmetrical about the first midline A1 and the second midline A2, or may be asymmetrical about each of the first midline A1 and the second midline A2, which is not limited. As illustrated in FIG. 7, in order to avoid the air-permeable holes **186**, four third push-point portions **174** (that is, two on each of the two second bosses **184**) provided on the fourth surface **185** are symmetrical about the first midline A1 and asymmetrical about the second midline A2.

[0140] By arranging the multiple third push-point portions **174** at intervals on the fourth surface **185**, the force can be uniformly applied to the fourth surface **185**, thereby facilitating demolding, and preventing the lower plastic member **10** from being damaged due to local uneven stress.

[0141] In an implementation, referring to FIG. 2, FIG. 3, and FIG. 7, the lower plastic member **10** further includes a first recessed platform **191** and a second recessed platform **194**. The first recessed platform **191** and the second recessed platform **194** are respectively disposed at two ends of the body **11** in the length direction of the body **11**, and are recessed from the first surface **111** and protrude from the second surface **112**. By providing the first recessed platform **191** and the second recessed platform **194**, a top surface of each of the first recessed platform **191** and the second recessed platform **194** protruding from the second surface **112** can be configured to be in contact with the cell assembly, so that there is a certain space between the body **11** and the cell assembly, thereby facilitating accommodation of structures such as a tab. Therefore, the space utilization rate of the energy storage apparatus is increased and the size of the energy storage apparatus is reduced. As illustrated in FIG. 7, the first recessed platform **191** may further have multiple first reinforcing

ribs **197** extending in the first direction X and arranged at intervals in the second direction Y. The second recessed platform **194** may further have multiple second reinforcing ribs **198** extending in the first direction X and arranged at intervals in the second direction Y. Each of the first reinforcing rib **197** and the second reinforcing rib **198** is configured to improve the structural strength of the lower plastic member **10**, which is not limited.

[0142] A bottom wall of the first recessed platform **191** recessed from the first surface **111** is a fifth surface **192**. A bottom wall of the second recessed platform **194** recessed from the first surface **111** is a sixth surface **195**. Each of the fifth surface **192** and the sixth surface **195** defines multiple leakage holes arranged at intervals. Each of the fifth surface **192** and the sixth surface **195** is provided with multiple third push-point portions **174** located in the intervals between the multiple leakage holes.

[0143] The multiple leakage holes include multiple first leakage holes **193** and multiple second leakage holes **196**. The fifth surface **192** defines multiple first leakage holes **193**. The sixth surface **195** defines multiple second leakage holes **196**. The multiple leakage holes allow electrolyte, which is sprayed onto the first surface **111** during injection, to flow into the cell assembly through injection holes. The multiple leakage holes may be defined at intervals in the second direction Y, and multiple corresponding third push-point portions **174** may be disposed at the intervals between the multiple leakage holes. One third push-point portion **174** may be disposed between two adjacent leakage holes. A third push-point portion **174** may be disposed outside the outermost leakage hole in the second direction Y. There are no limitations on this arrangement.

[0144] The multiple third push-point portions **174** disposed on the fifth surface **192** and the sixth surface **195** may be symmetrical about each of the first midline A1 and the second midline A2, or may be asymmetrical about each of the first midline A1 and the second midline A2, which is not limited. As illustrated in FIG. 7, in order to avoid the leakage holes, six third push-point portions **174** provided on the fifth surface **192** and six third push-point portions **174** provided on the sixth surface **195** are asymmetrical about the first midline A1, and symmetrical about the second midline A2.

[0145] By arranging the multiple third push-point portions **174** at intervals on each of the fifth surface **192** and the sixth surface **195**, the force can be uniformly applied to each of the fifth surface **192** and the sixth surface **195**, thereby facilitating demolding, and preventing the lower plastic member **10** from being damaged due to local uneven stress.

[0146] As illustrated in FIG. 7, the multiple first push-point portions **172**, the multiple second push-point portions **173**, and the multiple third push-point portions **174** may be arranged on the lower plastic member **10** as a whole relatively uniformly, so that the lower plastic member **10** as a whole can be stressed relatively uniformly, and the lower plastic member **10** is not easily damaged due to bending and deformation during demolding. For example, the multiple first push-point portions **172** may be evenly arranged around the terminal boss **14**, on both sides of the explosion-proof boss **18** opposite to each other in the first direction X, and the like. The multiple second push-point portions **173** may be evenly arranged on both sides of each of the four suction portions **12** opposite to each other in the first direction X, and on the third surface **182**. The multiple third push-point portions **174** may be evenly arranged on each of the fourth surface **185**, the fifth surface **192**, and the sixth surface **195**.

[0147] As illustrated in FIG. 3 and FIG. 6, the lower plastic member **10** further includes a gate portion **101**. The gate portion **101** is located at one side of the second surface **112** positioned facing away from the first surface **111**. The gate portion **101** is residually formed during injection molding of the lower plastic member **10**.

[0148] As illustrated in FIG. 4 and FIG. 5, the second mold **22** defines a pouring hole **225** extending through the second mold **22** in the third direction Z. The pouring hole **225** is in communication with the mold cavity. The pouring hole **225** allows the plastic liquid to flow, so that the plastic liquid can be injected into the mold cavity. After the plastic liquid is solidified and

molded, demolding is performed. After demolding, a pouring mark may remain at a position of the lower plastic member **10** corresponding to the pouring hole **225** to form a gate portion **101**. The gate portion **101** may be a protrusion, a groove, or other structures, which is not limited. The shape and size of the gate portion **101** are also not limited.

[0149] By forming the gate portion **101** at the side of the second surface **112** positioned facing away from the first surface **111**, the first mold **21** is configured for demolding by the push rod, and the second mold **22** is configured for pouring of the plastic liquid, so that the function of the first mold **21** is independent from the function of the second mold **22**. Therefore, misoperation, which is easily caused by performing the pouring of the plastic liquid and the demolding by the push rod on the same mold at the same time, can be avoided, and then adverse effects caused by the misoperation can be avoided.

[0150] In an embodiment, as illustrated in FIG. 6, a top surface of each of the explosion-proof boss **18**, the first recessed platform **191**, and the second recessed platform **194**, protruding from the second surface **112**, is provided with the gate portion **101**.

[0151] The gate portion **101** on the explosion-proof boss **18** may be disposed on the first boss **181** or the second boss **184**, and needs to avoid the vent hole **183** and the air-permeable hole **186**. The gate portions **101** disposed on the first recessed platform **191** and the second recessed platform **194** need to avoid the leakage hole.

[0152] The number of gate portions **101** may be one or more. When there is one gate portion **101**, the gate portions **101** may be disposed on any one of the explosion-proof bosses **18**, the first recessed platform **191**, and the second recessed platform **194**. When there are multiple gate portions **101**, the multiple gate portions **101** may be disposed on at least two of the explosion-proof bosses **18**, the first recessed platform **191**, and the second recessed platform **194**. Exemplarily, as illustrated in FIG. 6, each of the explosion-proof boss **18**, the first recessed platform **191**, and the second recessed platform **194** is provided with one gate portion **101**.

[0153] By providing the gate portion **101**, the injection molding process can be conveniently carried out, the interference with the demolding operation can be avoided, and the product yield can be ensured.

[0154] In the description of embodiments of the present disclosure, it should be noted that terms such as “center,” “upper,” “lower,” “left,” “right,” “vertical,” “horizontal,” “inner,” “outer,” and other indicators of orientation or positional relationships are based on the orientation or positional relationships described in the accompanying drawings. These terms are provided merely for the convenience of describing the present disclosure and simplifying the description and are not intended to indicate or imply that the referred apparatus or element must have a specific orientation, be constructed, or operate in a specific orientation. Therefore, these terms cannot be construed as limitations to the present disclosure.

[0155] What has been disclosed above is only a preferred embodiment of the present disclosure and cannot be used to limit the scope of the present disclosure. Those of ordinary skill in the art can understand that, all or part of the processes of realizing the above embodiments and the equivalent changes made according to the claims of the present disclosure, still fall within the scope of the present disclosure.

Claims

1. A lower plastic member for an end cover assembly of an energy storage apparatus, comprising: a body, having a first surface and a second surface positioned facing away from the first surface, wherein the first surface is configured to be positioned at one side of the body facing towards an upper cover; a suction portion, disposed on the first surface, wherein the suction portion comprises a plurality of first ribs and a plurality of second ribs, the plurality of first ribs are parallel to and spaced apart from one another, the plurality of second ribs are parallel to and spaced apart from one

another, the plurality of first ribs and the plurality of second ribs are disposed in a crisscross pattern, the plurality of first ribs and the plurality of second ribs cooperatively define a plurality of recesses, and each of the plurality of recesses is recessed from the first surface towards the second surface; and an explosion-proof boss, wherein the explosion-proof boss is disposed at a middle part of the body in a length direction of the body, the explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface, wherein the explosion-proof boss is configured to be aligned with an explosion-proof valve disposed on the upper cover; a midline extending in a width direction of the first surface is corresponding to the explosion-proof boss, and there is a spacing distance between the suction portion and the explosion-proof boss; an end surface of the plurality of first ribs configured to be positioned facing towards the upper cover is flush with the first surface, and an end surface of the plurality of second ribs configured to be positioned facing towards the upper cover is flush with the first surface; an orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm; an orthographic projection of each of the plurality of recesses on the first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm; and a depth of each of the plurality of recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm; and wherein the suction portion is implemented as a plurality of suction portions, and the plurality of suction portions are symmetrical about the midline extending in the width direction of the first surface.

2. The lower plastic member of claim 1, wherein an end portion of each of the plurality of first ribs and the plurality of second ribs, configured to be positioned facing towards the upper cover, has a pointed-prism shape.

3. The lower plastic member of claim 1, wherein at least two of the plurality of suction portions are disposed on a same side of the midline extending in the width direction of the first surface, and have different distances from the midline extending in the width direction of the first surface.

4. The lower plastic member of claim 1, wherein the first surface is provided with a code portion, the code portion has a preset shape, and the preset shape comprises at least one of a number, a letter, a bar code, or a two-dimensional code.

5. The lower plastic member of claim 4, wherein the code portion is implemented as a plurality of code portions arranged side by side.

6. The lower plastic member of claim 5, wherein any adjacent two of the plurality of code portions are spaced apart from each other.

7. The lower plastic member of claim 1, further comprising a terminal boss, wherein the terminal boss is connected to the body and protrudes from the first surface, and a snap-fit is provided at each of the two sides of the terminal boss in the width direction of the first surface.

8. The lower plastic member of claim 7, wherein the snap-fit comprises an extension portion and a protrusion, the extension portion has one end connected to the first surface and extends in a direction away from the second surface, and the protrusion is disposed at another end of the extension portion away from the body and protrudes from a surface of the extension portion positioned facing away from the terminal boss.

9. The lower plastic member of claim 8, further comprising a slider portion, wherein the slider portion is disposed on the first surface and located at one side of the extension portion positioned facing away from the terminal boss, and the slider portion is configured to allow a slider to slide on the first surface.

10. The lower plastic member of claim 9, further comprising an annular push portion, wherein the annular push portion is disposed on the first surface and surrounds the terminal boss, and the annular push portion is connected to or spaced apart from the slider portion.

11. The lower plastic member of claim 1, further comprising a plurality of first push-point portions, wherein the plurality of first push-point portions are disposed on the first surface and spaced apart from one another.

12. The lower plastic member of claim 11, wherein a bottom wall of the explosion-proof boss recessed from the first surface is a third surface, the third surface defines a vent hole, and each of the first surface and the third surface is provided with a plurality of second push-point portions spaced apart from one another.

13. The lower plastic member of claim 12, wherein a midline extending in a length direction of the first surface is a first midline, and the midline extending in the width direction of the first surface is a second midline; and the plurality of first push-point portions are symmetrical about each of the first midline and the second midline, and/or the plurality of second push-point portions are symmetrical about each of the first midline and the second midline.

14. The lower plastic member of claim 12, wherein the explosion-proof boss comprises a first boss and two second bosses, the first boss is disposed at a middle part of the first surface in the width direction of the first surface, two ends of the first boss in the width direction of the first surface are connected to the two second bosses, respectively, a depth of the first boss recessed relative to the first surface is less than a depth of each second boss recessed relative to the first surface, a bottom wall of the first boss is the third surface, a bottom wall of the second boss recessed from the first surface is a fourth surface, the fourth surface defines a plurality of air-permeable holes arranged at intervals, and the fourth surface is provided with a plurality of third push-point portions located in the intervals between the plurality of air-permeable holes.

15. The lower plastic member of claim 14, further comprising a first recessed platform and a second recessed platform, wherein the first recessed platform and the second recessed platform are respectively disposed at two ends of the body in the length direction of the body, and are recessed from the first surface and protrude from the second surface, a bottom wall of the first recessed platform recessed from the first surface is a fifth surface, a bottom wall of the second recessed platform recessed from the first surface is a sixth surface, each of the fifth surface and the sixth surface defines a plurality of leakage holes arranged at intervals, and each of the fifth surface and the sixth surface is provided with a plurality of third push-point portions located in the intervals between the plurality of leakage holes.

16. The lower plastic member of claim 11, wherein each of the plurality of first push-point portions is crater-shaped, a middle part of each of the plurality of first push-point portions is recessed relative to the first surface, and a peripheral edge of each of the plurality of first push-point portions protrudes relative to the first surface.

17. The lower plastic member of claim 1, further comprising a gate portion, wherein the gate portion is located at one side of the second surface positioned facing away from the first surface, and the gate portion is residually formed during injection molding of the lower plastic member.

18. The lower plastic member of claim 17, further comprising a first recessed platform, and a second recessed platform, wherein the explosion-proof boss is disposed at a middle part of the body in the length direction of the body, the first recessed platform and the second recessed platform are respectively disposed at two ends of the body in the length direction of the body, each of the explosion-proof boss, the first recessed platform, and the second recessed platform is recessed from the first surface and protrudes from the second surface, and a top surface of each of the explosion-proof boss, the first recessed platform, and the second recessed platform, protruding from the second surface, is provided with the gate portion.

19. An end cover assembly comprising: an upper cover and a lower plastic member, wherein the lower plastic member comprises: a body, having a first surface and a second surface positioned facing away from the first surface, wherein the first surface of the lower plastic member is positioned facing towards the upper cover; a suction portion, disposed on the first surface, wherein the suction portion comprises a plurality of first ribs and a plurality of second ribs, the plurality of first ribs are parallel to and spaced apart from one another, the plurality of second ribs are parallel to and spaced apart from one another, the plurality of first ribs and the plurality of second ribs are disposed in a crisscross pattern, the plurality of first ribs and the plurality of second ribs

cooperatively define a plurality of recesses, and each of the plurality of recesses is recessed from the first surface towards the second surface; and an explosion-proof boss, wherein the explosion-proof boss is disposed at a middle part of the body in a length direction of the body, the explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface; wherein the explosion-proof boss is aligned with an explosion-proof valve disposed on the upper cover; wherein a midline extending in a width direction of the first surface is corresponding to the explosion-proof boss, and there is a spacing distance between the suction portion and the explosion-proof boss; wherein an end surface of the plurality of first ribs positioned facing towards the upper cover is flush with the first surface, and an end surface of the plurality of second ribs positioned facing towards the upper cover is flush with the first surface; wherein an orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm; an orthographic projection of each of the plurality of recesses on the first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm; and a depth of each of the plurality of recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm; and wherein the suction portion is implemented as a plurality of suction portions, and the plurality of suction portions are symmetrical about the midline extending in the width direction of the first surface.

20. A power consuming device comprising: a power consuming apparatus and an energy storage apparatus, wherein the energy storage apparatus is configured to supply power to the power consuming apparatus, and the energy storage apparatus comprises a housing, a cell assembly, and an end cover assembly, wherein the cell assembly is disposed in the housing, and the end cover assembly is connected to the housing; and the end cover assembly comprises an upper cover and a lower plastic member, wherein the lower plastic member comprises: a body, having a first surface and a second surface positioned facing away from the first surface, wherein the first surface of the lower plastic member is positioned facing towards the upper cover; a suction portion, disposed on the first surface, wherein the suction portion comprises a plurality of first ribs and a plurality of second ribs, the plurality of first ribs are parallel to and spaced apart from one another, the plurality of second ribs are parallel to and spaced apart from one another, the plurality of first ribs and the plurality of second ribs are disposed in a crisscross pattern, the plurality of first ribs and the plurality of second ribs cooperatively define a plurality of recesses, and each of the plurality of recesses is recessed from the first surface towards the second surface; and an explosion-proof boss, wherein the explosion-proof boss is disposed at a middle part of the body in a length direction of the body, the explosion-proof boss is recessed relative to the first surface and protrudes relative to the second surface; wherein the explosion-proof boss is aligned with an explosion-proof valve disposed on the upper cover; wherein a midline extending in a width direction of the first surface is corresponding to the explosion-proof boss, and there is a spacing distance between the suction portion and the explosion-proof boss; wherein an end surface of the plurality of first ribs positioned facing towards the upper cover is flush with the first surface, and an end surface of the plurality of second ribs positioned facing towards the upper cover is flush with the first surface; wherein an orthographic projection of the suction portion on the first surface is circular and has a diameter ranging from 5 mm to 10 mm; an orthographic projection of each of the plurality of recesses on the first surface is rectangular, and has a length ranging from 0.3 mm to 0.6 mm and a width ranging from 0.3 mm to 0.6 mm; and a depth of each of the plurality of recesses recessed from the first surface ranges from 0.2 mm to 0.3 mm; and wherein the suction portion is implemented as a plurality of suction portions, and the plurality of suction portions are symmetrical about the midline extending in the width direction of the first surface.
